

Reference	Qty	Value	Footprint	Datasheet
A1	1	RaspberryPi_Pico	Module:RaspberryPi_Pico_Common_Unspecified	https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf Note that any Pico 2 (2350) device could be used, as long as pin compatible
C3	1	100nF	Capacitor_THT:C_Disc_D4.3mm_W1.9mm_P5.00mm	
C6	1	100pF	Capacitor_THT:C_Disc_D4.3mm_W1.9mm_P5.00mm	
D1,D2,D5,D6	4	5.1V	Diode_THT:D_DO-35_SOD27_P7.62mm_Horizontal	
D3,D4	2	1N4148	Diode_THT:D_DO-35_SOD27_P7.62mm_Horizontal	
R1,R2,R11,R12,R13,R14,R15	7	220	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	
R3,R4,R5	3	10k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	
R6,R7,R16,R17	4	4,7k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	
R8,R9	2	75	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	
R10	1	150	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal	
T1,T2	2	78601_3JC	78601_3JC:XFMR_78601_3JC	1:1 transformers
U3	1	LM319	Package_DIP:DIP-14_W7.62mm	http://www.ti.com/lit/ds/symlink/lm319-n.pdf
Optional				
D7,D8,D9,D10,D11	5	LED	LED_THT:LED_D5.0mm	
D12,D13,D14,D15,D16	5	WS2812D	LED_D5.0mm-4_RGB	https://jlcpcb.com/partdetail/WS2812D-255/C4154874
Connectors				
J1	1	Conn_01x06_Pin	Connector_PinHeader_2.54mm:PinHeader_1x06_P2.54mm_Vertical	
J2	1	Conn_01x03_Pin	Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical	
J3,J6	2	Conn_01x05_Pin	Connector_PinHeader_2.54mm:PinHeader_1x05_P2.54mm_Vertical	
J4	1	Conn_01x04_MountingPin	JST_SH_SM04B-SRSS-TB_1x04-1MP_P1.00mm_Horizontal	
J5	1	Screw_Terminal_01x04	Connector_Molex:Molex_KK-254_AE-6410-04A_1x04_P2.54mm_Vertical	To connect the HP-IL cables
JP1	1	Conn_02x20_Odd_Even	Connector_PinHeader_2.54mm:PinHeader_2x20_P2.54mm_Vertical	
JP2	1	Conn_01x08_Socket	Connector_PinHeader_2.54mm:PinHeader_2x04_P2.54mm_Vertical	